Course 6014B | Semester VI | Theory | 4 Credits

Sustainable Development

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6014B as Sustainable Development in Semester VI for Civil Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Hyderabad’s courses on Strategies for Sustainable Design and UN SDGs, and polytechnic sustainable development curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

> **Source treatment:** Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Sustainability assessments, environmental impact reports, carbon footprint calculations and development policy applications must be based on authorised environmental data, certified assessment methodologies, current legislation and competent professional review.

> **Verified source note:** Verified sources support sustainability definitions, pillars (environmental, social, economic), the 17 UN SDGs, circular economy, design for sustainability, EIA and LCA. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Sustainable Development studies the principles and practices required to meet current human needs without compromising the ability of future generations to meet theirs. It develops the ability to analyze the three pillars of sustainability (environmental, social, economic), apply the 17 UN Sustainable Development Goals (SDGs), evaluate resource management through circular economy principles, and conduct sustainability assessments while respecting the global and local boundaries of development.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□ □□□□□□□□.

Prerequisite foundation

Environmental studies, basic social sciences, elementary economics, engineering mathematics and global awareness.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain sustainable development fundamentals, the three pillars and the transition from MDGs to SDGs.
2. Explain the 17 UN Sustainable Development Goals (SDGs), their targets and systemic perspectives.
3. Explain sustainable resource management, circular economy principles and design for sustainability.
4. Explain sustainability assessment tools, including Environmental Impact Assessment (EIA) and Life Cycle Analysis (LCA).

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□□ □□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□□□. Define, explain, apply □□□□□□ action words □□□□□□□□□□□□□.

Official Course Outcomes		
CO	**Official outcome**	**Bloom level**
CO1	Explain the concepts, pillars and historic evolution of sustainable development.	Understand
CO2	Explain the 17 UN Sustainable Development Goals and their interconnectedness.	Understand
CO3	Explain circular economy principles and strategies for sustainable resource management.	Understand
CO4	Apply sustainability assessment methods like EIA and LCA for development projects.	Apply

Official CO-PO articulation matrix							
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7–PO11**
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Sustainability Principles and Evolution	Definition and characteristics of sustainability; the three pillars (environmental, social, economic); Brundtland report ('Our Common Future'); historic origins; transition from MDGs to SDGs; Agenda 2030.	Approx. 14 hours	CO1	Module 1
2	The 17 UN Sustainable Development Goals	Overview of the 17 SDGs (No Poverty to Partnerships); systemic perspectives and interconnectedness; people, ecological and spiritual categories; SDGs and socio-ecological systems; global and national reports.	Approx. 16 hours	CO2	Module 2
3	Sustainable Resource and Systemic Design	Circular economy principles (reduce, reuse, recycle); design for sustainability (D4S); systemic design thinking; sustainable building materials; energy efficiency in the built environment; vernacular design.	Approx. 16 hours	CO3	Module 3
4	Sustainability Assessment and Policy	Environmental Impact Assessment (EIA); Life Cycle Analysis (LCA) - principles and stages; ecological and carbon footprints; international conventions and laws; emerging technologies for SD; net-zero targets.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Sustainability Principles and Evolution

Definition and characteristics of sustainability; the three pillars (environmental, social, economic); Brundtland report ('Our Common Future'); historic origins; transition from MDGs to SDGs; Agenda 2030.

CO1 · Approx. 14 hours

The 17 UN Sustainable Development Goals

Overview of the 17 SDGs (No Poverty to Partnerships); systemic perspectives and interconnectedness; people, ecological and spiritual categories; SDGs and socio-ecological systems; global and national reports.

CO2 · Approx. 16 hours

Sustainable Resource and Systemic Design

Circular economy principles (reduce, reuse, recycle); design for sustainability (D4S); systemic design thinking; sustainable building materials; energy efficiency in the built environment; vernacular design.

CO3 · Approx. 16 hours

Sustainability Assessment and Policy

Environmental Impact Assessment (EIA); Life Cycle Analysis (LCA) - principles and stages; ecological and carbon footprints; international conventions and laws; emerging technologies for SD; net-zero targets.

CO4 · Approx. 14 hours

MODULE 1

Sustainability Principles and Evolution

Definition and characteristics of sustainability; the three pillars (environmental, social, economic); Brundtland report ('Our Common Future'); historic origins; transition from MDGs to SDGs; Agenda 2030.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

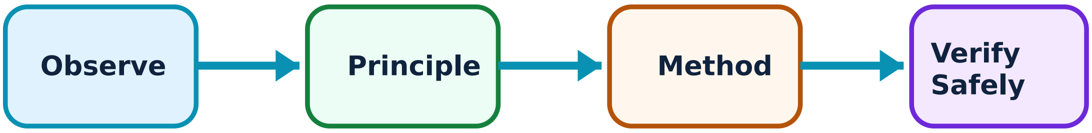
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Sustainability Principles and Evolution: identify the system, state its principle, follow the correct method and verify the result safely.

1. Concepts and pillars of sustainability−

Sustainability balances environmental protection, social equity and economic viability. The Brundtland Report defined it as meeting present needs without compromising the future. Understanding these interconnected pillars is essential for designing resilient and equitable systems.

Study and application method

List the three pillars of sustainability and explain the concept of 'intergenerational equity' as defined in the Brundtland Report.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the three pillars of sustainability and explain the concept of 'intergenerational equity' as defined in the Brundtland Report.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Sustainability is a dynamic and context-dependent concept. Do not assume that a solution for one region or sector is universally applicable without local social and environmental analysis.

2. Historic evolution and Agenda 2030−

The global sustainability agenda evolved from the Millennium Development Goals (MDGs) to the more comprehensive Sustainable Development Goals (SDGs). Agenda 2030 provides the current global framework for peace and prosperity for people and the planet.

Study and application method

Describe the major differences between MDGs and SDGs; explain the role of the United Nations in promoting global sustainable development.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the major differences between MDGs and SDGs; explain the role of the United Nations in promoting global sustainable development.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Policy goals are aspirations. Achieving them requires coordinated local, national and international action and continuous monitoring of indicators.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

The 17 UN Sustainable Development Goals

Overview of the 17 SDGs (No Poverty to Partnerships); systemic perspectives and interconnectedness; people, ecological and spiritual categories; SDGs and socio-ecological systems; global and national reports.

Approx. 16 hours **CO2** **Understand · Apply**

Learning goals

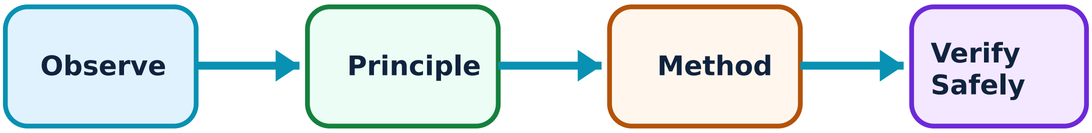
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for The 17 UN Sustainable Development Goals: identify the system, state its principle, follow the correct method and verify the result safely.

1. SDGs and their interconnectedness−

The 17 SDGs cover diverse areas like poverty, health, education, energy, climate and justice. They are interconnected—progress in one goal (e.g., SDG 7: Clean Energy) often supports others (e.g., SDG 13: Climate Action). Systemic mapping helps identify these synergies.

Study and application method

Select three SDGs and explain how they are interconnected in a specific local context (e.g., a rural village or an urban slum).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Select three SDGs and explain how they are interconnected in a specific local context (e.g., a rural village or an urban slum).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Prioritising one SDG without considering others can lead to unintended negative consequences. Use a systemic approach to avoid 'siloed' development interventions.

2. Implementation and assessment−

SDG implementation involves financing, capacity building and policy alignment. In India, NITI Aayog monitors progress through the SDG India Index. Global and national reports track indicators to ensure accountability and identify areas needing urgent attention.

Study and application method

Identify the nodal agency for SDG implementation in India and list four indicators used to measure progress toward SDG 6 (Clean Water and Sanitation).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify the nodal agency for SDG implementation in India and list four indicators used to measure progress toward SDG 6 (Clean Water and Sanitation).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Indicator data may be incomplete or subject to reporting lags. Use authorised national and international databases for official sustainability assessments.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Sustainable Resource and Systemic Design

Circular economy principles (reduce, reuse, recycle); design for sustainability (D4S); systemic design thinking; sustainable building materials; energy efficiency in the built environment; vernacular design.

Approx. 16 hours | **CO3** | **Understand · Apply**

Learning goals

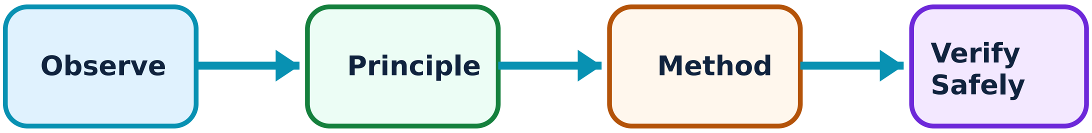
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Sustainable Resource and Systemic Design: identify the system, state its principle, follow the correct method and verify the result safely.

1. Circular economy and D4S−

A circular economy moves away from 'take-make-waste' to a closed-loop system where resources are kept in use for as long as possible. Design for Sustainability (D4S) integrates environmental and social considerations into the entire product or system life cycle.

Study and application method

Compare a linear economy with a circular economy using a conceptual diagram; list three strategies for 'design for sustainability' in construction.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare a linear economy with a circular economy using a conceptual diagram; list three strategies for 'design for sustainability' in construction.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Circular systems require robust infrastructure and behavioral change. Do not assume recycling alone constitutes a circular economy without considering reduction and reuse.

2. Systemic design and built environment−

Systemic design identifies alternatives to consumption-heavy lifestyles. In the built environment, this includes using sustainable materials, improving energy efficiency and learning from climate-responsive vernacular architecture to reduce environmental impact.

Study and application method

Describe how vernacular architecture in your region demonstrates sustainable design principles; identify two sustainable alternatives to conventional building materials.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe how vernacular architecture in your region demonstrates sustainable design principles; identify two sustainable alternatives to conventional building materials.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Technological fixes (e.g., high-efficiency HVAC) must be balanced with passive design and lifestyle changes for true sustainability.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Sustainability Assessment and Policy

Environmental Impact Assessment (EIA); Life Cycle Analysis (LCA) - principles and stages; ecological and carbon footprints; international conventions and laws; emerging technologies for SD; net-zero targets.

Approx. 14 hours **CO4** **Understand · Apply**

Learning goals

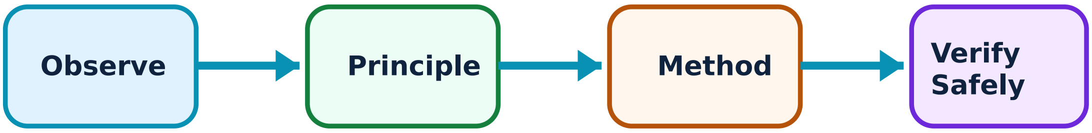
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Sustainability Assessment and Policy: identify the system, state its principle, follow the correct method and verify the result safely.

1. Assessment tools: EIA and LCA−

EIA predicts the environmental consequences of a project before implementation. LCA evaluates the impact of a product or system throughout its life ('cradle to grave'). Both are essential for evidence-based decision-making in sustainable development.

Study and application method

Outline the major stages of a Life Cycle Analysis (LCA) for a building component; explain the purpose of a 'scoping' meeting in an EIA process.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Outline the major stages of a Life Cycle Analysis (LCA) for a building component; explain the purpose of a 'scoping' meeting in an EIA process.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: EIA and LCA are technical assessments. They must be conducted by accredited professionals using verified data and authorised methodologies.

2. Climate policy and net-zero−

International conventions (like the Paris Agreement) and national laws provide the regulatory framework for climate action. Net-zero targets aim to balance greenhouse gas emissions with removals, requiring rapid adoption of emerging green technologies.

Study and application method

Define 'net-zero emissions' and list three international climate agreements that India is a party to.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'net-zero emissions' and list three international climate agreements that India is a party to.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Net-zero targets involve long-term projections and technological uncertainties. All climate actions must comply with current national and international environmental laws.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Sustainability Principles and Evolution

Use the concept flow to revise observation, principle, method and verification.

Module 2: The 17 UN Sustainable Development Goals

Use the concept flow to revise observation, principle, method and verification.

Module 3: Sustainable Resource and Systemic Design

Use the concept flow to revise observation, principle, method and verification.

Module 4: Sustainability Assessment and Policy

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Map the interconnectedness of five UN SDGs for a hypothetical sustainable-city project.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the flow of materials in a circular economy for a specific industry (e.g., textiles or construction).

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify three vernacular building techniques in your region and explain their sustainability benefits.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Outline the steps for a Life Cycle Analysis (LCA) of a consumer product like a smartphone or a solar panel.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare a summary of India's current status and targets for three specific Sustainable Development Goals.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Sustainability Principles and Evolution

Explain Concepts and pillars of sustainability and state one correct method, application or precaution.−

Sustainability balances environmental protection, social equity and economic viability. The Brundtland Report defined it as meeting present needs without compromising the future. Understanding these interconnected pillars is essential for designing resilient and equitable systems.

Answer framework: List the three pillars of sustainability and explain the concept of 'intergenerational equity' as defined in the Brundtland Report.

Important point: Sustainability is a dynamic and context-dependent concept. Do not assume that a solution for one region or sector is universally applicable without local social and environmental analysis.

Explain Historic evolution and Agenda 2030 and state one correct method, application or precaution.−

The global sustainability agenda evolved from the Millennium Development Goals (MDGs) to the more comprehensive Sustainable Development Goals (SDGs). Agenda 2030 provides the current global framework for peace and prosperity for people and the planet.

Answer framework: Describe the major differences between MDGs and SDGs; explain the role of the United Nations in promoting global sustainable development.

Important point: Policy goals are aspirations. Achieving them requires coordinated local, national and international action and continuous monitoring of indicators.

Module 2 — The 17 UN Sustainable Development Goals

Explain SDGs and their interconnectedness and state one correct method, application or precaution.−

The 17 SDGs cover diverse areas like poverty, health, education, energy, climate and justice. They are interconnected—progress in one goal (e.g., SDG 7: Clean Energy) often supports others (e.g., SDG 13: Climate Action). Systemic mapping helps identify these synergies.

Answer framework: Select three SDGs and explain how they are interconnected in a specific local context (e.g., a rural village or an urban slum).

Important point: Prioritising one SDG without considering others can lead to unintended negative consequences. Use a systemic approach to avoid 'siloed' development interventions.

Explain Implementation and assessment and state one correct method, application or precaution.−

SDG implementation involves financing, capacity building and policy alignment. In India, NITI Aayog monitors progress through the SDG India Index. Global and national reports track indicators to ensure accountability and identify areas needing urgent attention.

Answer framework: Identify the nodal agency for SDG implementation in India and list four indicators used to measure progress toward SDG 6 (Clean Water and Sanitation).

Important point: Indicator data may be incomplete or subject to reporting lags. Use authorised national and international databases for official sustainability assessments.

Module 3 — Sustainable Resource and Systemic Design

Explain Circular economy and D4S and state one correct method, application or precaution.−

A circular economy moves away from 'take-make-waste' to a closed-loop system where resources are kept in use for as long as possible. Design for Sustainability (D4S) integrates environmental and social considerations into the entire product or system life cycle.

Answer framework: Compare a linear economy with a circular economy using a conceptual diagram; list three strategies for 'design for sustainability' in construction.

Important point: Circular systems require robust infrastructure and behavioral change. Do not assume recycling alone constitutes a circular economy without considering reduction and reuse.

Explain Systemic design and built environment and state one correct method, application or precaution.−

Systemic design identifies alternatives to consumption-heavy lifestyles. In the built environment, this includes using sustainable materials, improving energy efficiency and learning from climate-responsive vernacular architecture to reduce environmental impact.

Answer framework: Describe how vernacular architecture in your region demonstrates sustainable design principles; identify two sustainable alternatives to conventional building materials.

Important point: Technological fixes (e.g., high-efficiency HVAC) must be balanced with passive design and lifestyle changes for true sustainability.

Module 4 — Sustainability Assessment and Policy

Explain Assessment tools: EIA and LCA and state one correct method, application or precaution.−

EIA predicts the environmental consequences of a project before implementation. LCA evaluates the impact of a product or system throughout its life ('cradle to grave'). Both are essential for evidence-based decision-making in sustainable development.

Answer framework: Outline the major stages of a Life Cycle Analysis (LCA) for a building component; explain the purpose of a 'scoping' meeting in an EIA process.

Important point: EIA and LCA are technical assessments. They must be conducted by accredited professionals using verified data and authorised methodologies.

Explain Climate policy and net-zero and state one correct method, application or precaution.−

International conventions (like the Paris Agreement) and national laws provide the regulatory framework for climate action. Net-zero targets aim to balance greenhouse gas emissions with removals, requiring rapid adoption of emerging green technologies.

Answer framework: Define 'net-zero emissions' and list three international climate agreements that India is a party to.

Important point: Net-zero targets involve long-term projections and technological uncertainties. All climate actions must comply with current national and international environmental laws.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Sustainability Principles and Evolution.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: The 17 UN Sustainable Development Goals.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Sustainable Resource and Systemic Design.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Sustainability Assessment and Policy.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.

2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Definition and characteristics of sustainability; the three pillars (environmental, social, economic); Brundtland report ('Our Common Future'); historic origins; transition from MDGs to SDGs; Agenda 2030..
2. Develop a complete answer or practical plan for: Overview of the 17 SDGs (No Poverty to Partnerships); systemic perspectives and interconnectedness; people, ecological and spiritual categories; SDGs and socio-ecological systems; global and national reports..
3. Develop a complete answer or practical plan for: Circular economy principles (reduce, reuse, recycle); design for sustainability (D4S); systemic design thinking; sustainable building materials; energy efficiency in the built environment; vernacular design..
4. Develop a complete answer or practical plan for: Environmental Impact Assessment (EIA); Life Cycle Analysis (LCA) - principles and stages; ecological and carbon footprints; international conventions and laws; emerging technologies for SD; net-zero targets..

LAST-MILE REVISION

Quick Revision

Module 1: Sustainability Principles and Evolution

Definition and characteristics of sustainability; the three pillars (environmental, social, economic); Brundtland report ('Our Common Future'); historic origins; transition from MDGs to SDGs; Agenda 2030.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: The 17 UN Sustainable Development Goals

Overview of the 17 SDGs (No Poverty to Partnerships); systemic perspectives and interconnectedness; people, ecological and spiritual categories; SDGs and socio-ecological systems; global and national reports.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Sustainable Resource and Systemic Design

Circular economy principles (reduce, reuse, recycle); design for sustainability (D4S); systemic design thinking; sustainable building materials; energy efficiency in the built environment; vernacular design.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Sustainability Assessment and Policy

Environmental Impact Assessment (EIA); Life Cycle Analysis (LCA) - principles and stages; ecological and carbon footprints; international conventions and laws; emerging technologies for SD; net-zero targets.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Concepts and pillars of sustainability	Sustainability balances environmental protection, social equity and economic viability.
Historic evolution and Agenda 2030	The global sustainability agenda evolved from the Millennium Development Goals (MDGs) to the more comprehensive Sustainable Development Goals (SDGs).
SDGs and their interconnectedness	The 17 SDGs cover diverse areas like poverty, health, education, energy, climate and justice.
Implementation and assessment	SDG implementation involves financing, capacity building and policy alignment.
Circular economy and D4S	A circular economy moves away from 'take-make-waste' to a closed-loop system where resources are kept in use for as long as possible.
Systemic design and built environment	Systemic design identifies alternatives to consumption-heavy lifestyles.
Assessment tools: EIA and LCA	EIA predicts the environmental consequences of a project before implementation.
Climate policy and net-zero	International conventions (like the Paris Agreement) and national laws provide the regulatory framework for climate action.

LEARNING RESOURCES

References

Recommended sustainable-development references

1. Jeffrey Sachs, The Age of Sustainable Development.
2. UN, Transforming our world: the 2030 Agenda for Sustainable Development.
3. Shiva Ji, Strategies for Sustainable Design.
4. NITI Aayog, SDG India Index Reports.

Approved public sustainable-development sources

- https://nptel.ac.in/courses/124106157
- https://nptel.ac.in/courses/109106200

These sources provide the verified study basis while official Course 6014B detail is unavailable; they do not replace official environmental impact reports, authorised sustainability certifications, certified LCA data or legal policy mandates.